Physical Chemistry Cume

Compare the pH of the following solutions:

(1) 0.100 M $HC_2H_3O_2$

(2) 0.100 M $NaC_2H_3O_2$

(3) A buffer solution prepared by adding 0.500 mol of $NaC_2H_3O_2$ to 1 dm^3 of 0.10 M $HC_2H_3O_2$

(4) The buffer solution after the addition of 0.001 mol NaOH

For the reaction

$$HC_2H_3O_2(aq) = H^+ (aq) + C_2H_3O_2^- (aq)$$

$$K_a = 1.76 \times 10^{-5}$$

K_a is the acid ionization constant
K_b is the base ionization constant
K_w is the water ionization constant

For the salt of a strong base and weak acid the hydrolysis constant is

$$K_h = K_w/K_a$$

For the salt of a strong acid and a weak base

$$K_h = K_w / K_b$$

For the salt of a weak base and a weak acid

$$K_h = K_w/K_aK_b$$